Original Article

A new approach for Baltic Dry Index forecasting based on empirical mode decomposition and neural networks

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Abstract In this article, a method based on empirical mode decomposition (EMD) and artificial neural networks (ANN) is developed for Baltic Dry Index (BDI) forecasting. The original BDI series is decomposed into several independent intrinsic mode functions (IMFs) using EMD first. Then the IMFs are composed into three components: short-term fluctuations, effect of extreme events and long-term trend. On the basis of results of decomposition and composition, ANN is used to model each IMF and composed component. Results show that the proposed EMD-ANN method outperforms ANN and VAR. The EMD-based method thus provides a useful technique for dry bulk market analysis and forecasting.

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Keywords: dry bulk shipping market; empirical mode decomposition; artificial neural networks; forecasting; Baltic Dry Index (BDI)

Introduction

The dry bulk shipping market is the major component of international shipping market and it has the characteristics of seasonality, cyclicality, high volatility and capital intensiveness. Owing to the magnitude of investments required and the frequent fluctuations of freight rates, the forecasting of freight rates attracts much attention both from scholars and business practitioners. However, due to the

complexity of the bulk shipping market and the non-stationary and non-linear nature of freight rates series (Goulielmos and Psifia, 2009), their accurate prediction presents researchers with certain challenges (Goulielmos and Psifia, 2013).

In the past decades, econometric and statistical methods, such as ARIMA, VAR (vector auto-regression), GARCH and VECM (vector error correction model) models have been used in the analysis and forecasting of the shipping market. For example, Kavussanos and Nomikos (2003) found that VECM generated the most accurate forecasts of spot prices, but not of future prices. Batchelor *et al* (2007) compared ARIMA, VAR and VECM in predicting spot and forward freight rates. Their results demonstrated that ARIMA provided better forecasts of forward prices than spot prices, and VAR, VECM slightly outperformed ARIMA in predicting spot prices. Jing *et al* (2008) investigated the characteristics of volatility in dry freight rates by GARCH model, and the fficiency of GARCH model was verified. Finally, Chen *et al* (2012) applied ARIMA and VAR to predict freight rates of several dry bulk routes, and their results indicated that VAR performed better on the out-of-sample forecast against ARIMA.

Usually, the aforementioned econometric methods can give decent analysis and forecasting results, but only when the studied time series are linear. However, in reality, as mentioned earlier, in dry bulk shipping market, a great deal of non-linearity exists, which makes forecasting a challenge task (Goulielmos and Psifia, 2009). As stated by Stopford (2009, p. 697), *maritime forecasting in particular has a poor reputation*. One of the main reasons is that it is difficult for traditional econometric and statistical methods to capture the non-linear characteristics hidden in dry bulk freight series.

In this regard, a number of non-linear and artificial intelligence (AI) methods, such as artificial neural networks (ANN), support vector machines (SVM) and non-linear regression have been used. For example, Leonova and Nikolov (2012) proposed a model based on wavelet and neural network for the prediction of dry bulk freight rates. Bulut *et al* (2012) developed a vector autoregressive fuzzy integrated logical forecasting model for time charter rates. Duru *et al* (2012) proposed a fuzzy-DELPHI adjustment method to improve the accuracy of statistical forecasts in the dry bulk shipping index. Finally, Goulielmos and Psifia (2009) developed a non-linear method in which the non-linear dynamic and chaotic deterministic modeling theory was tested. These studies suggest that the AI-based methods possess considerable advantages over the traditional econometric and statistical methods. ANN is a significant branch of AI. The advantages of ANN are its suitability for non-linear relationships with arbitrary complexity, strong robustness and fault tolerance, fast arithmetic speed, and so on. However, the AI-based method also suffers from certain disadvantages. For example, ANN often suffers from local minima, while other methods such as SVM are sensitive to parameter selection.

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Empirical mode decomposition (EMD) is a non-linear, non-stationary data analysis method developed by Huang *et al* (1998). By EMD, a complicated signal can be decomposed into several independent intrinsic mode functions (IMFs) with simpler frequency components. Also, it can reveal the hidden characteristics and the trend of time series, thus serving as a highly effective tool in forecasting. As a result, EMD has been widely used in many fields, for example, machinery fault diagnosis (Qi *et al,* 2007), sea wave data analysis (Huang *et al,* 1999), medical science (Rojas *et al,* 2012), financial market forecasting (Guhathakurta *et al,* 2008), exchange rates (Premanode and Toumazou, 2013), metro passenger flow (Wei and Chen, 2012) and so on. In fault diagnosis, EMD performs well in extracting fault messages, reducing interference of vibration signal by external factors, diminishing energy diffusion and leakage, and improving signal accuracy. For the image of physiological signals, decomposition can reduce the boundary distortion of image. Furthermore, it works well in the forecasting of crude oil prices (Yu *et al*, 2008), exchange rates (Premanode and Toumazou, 2013), metro passenger flows (Wei and Chen, 2012) and so on.

Bulk shipping is similar to financial markets in many aspects, such as non-stationary and non-linear price series. The efficiency of EMD in financial market forecasting has been proven, and existing studies can provide meaningful reference. Zeng and Qu (2014) proved the efficiency of EMD in analyzing the dry bulk market. On the basis of this, we thus propose a method based on EMD and ANN to predict the Baltic Dry Index (BDI). This is a composite index, published by London's Baltic Exchange, and regarded as the 'barometer' of the dry bulk shipping market. In this article, the original BDI series is decomposed into several independent IMFs using EMD first, and the IMFs are composed into three components, namely, short-term fluctuations caused by normal market activities, the effect of extreme events and a long-term trend. A three-layer feed-forward neural network (FNN) is used to model each of the IMFs and composed components (CCs), and the prediction results of all IMFs and CCs are combined to formulate the prediction output for the initial BDI series. The proposed method is compared with existing ANN methods and traditional econometric models such as VAR.

EMD is a non-linear, non-stationary data analysis method, and ANN is suitable for analysis of non-linear relationships with arbitrary complexity. The integration of EMD and ANN is a potentially efficient method to tackle the non-linear and non-stationary nature of BDI. The method, based on EMD and ANN, has been proved to be efficient in forecasting crude oil prices (Yu *et al*, 2008), tourism demand (Chen *et al,* 2012) and so on. This article extends existing research by incorporating a composition process and improving the shifting process of EMD. Furthermore, to authors' knowledge, existing literature on BDI forecasting has not adopted EMD yet. This study fills this gap and provides a potentially effective approach for the analysis and prediction of dry bulk markets.

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The rest of the article is organized as follows. The next section illustrates the methodology formulation based on EMD and ANN. The empirical experiments are provided in penultimate section. Finally, the conclusions can be found in the final section.

Methodology Formulation

In this section, the overall process of formulating the forecasting methodology for the BDI series, based on EMD and ANN, is presented. First, a brief description of the EMD technique is provided, and the improved EMD for the BDI series analysis is developed. Then the forecasting methodology is proposed and the overall steps of this method are presented.

The improved EMD for BDI series analysis

The basic principle of EMD is to decompose a time series into several IMFs. The IMFs must satisfy two conditions: (i) The number of extrema (including maxima and minima) must be equal to the number of zero crossings throughout the whole time series, or differ at most by one; (ii) at any point, the mean value of local maximum envelope and local minimum envelope must be zero.

An IMF is nearly a periodic function, and represents a simple oscillatory mode comparing with the general harmonic function. Usually, the IMFs are obtained through a 'sifting process', which is described as follows:

(1) Identify all the local maxima and minima of the time series $x(t)$.
(2) Connect all the local maxima by the cubic spline, and obtain the upper envelope $h(t)$; similarly, use the local minima to obtain the lower envelope $l(t)$.
(3) Calculate the mean value of the upper and lower envelopes $m(t)$, $m(t) = ((1)/(2))[h(t) + l(t)]$.
(4) Obtain a new series $a(t)$, $a(t) = x(t) - m(t)$.
(5) If $a(t)$ satisfies the characteristics of IMFs, $a(t)$ is regarded as an IMF, replace the $x(t)$ in Step (1) with $r(t) = x(t) - a(t)$; Otherwise, replace the $x(t)$ in Step (1) with $a(t)$.
(6) Repeat steps (1–5) until the residue component $r(t)$ becomes a monotonic function, which is the stopping criterion suggested by Huang *et al* (1999).

At the end of the sifting process, the original time series $x(t)$ is decomposed into several IMFs $a_i(t)$ and a residue $r(t)$, which can be denoted as follows:

$$x(t) = \sum_{i=1}^{n} a_i(t) + r(t) \quad (1)$$

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EMD has been used as an effective method to extract signals from data generated in non-linear and non-stationary time series. The method is easy to be understood and implemented, and fluctuations within a time series can be selected from the time series automatically and adaptively. However, recognizing that the original economic nature of the BDI time series may be lost to a large extent by EMD, an improved EMD (EEMD) is needed so as to address this problem. The basic idea is to classify and reconstitute the IMFs based on the decomposition results of EMD. The process is shown in Figure 1 and illustrated as follows:

(1) Sequence $a_i(t)$ according to the frequency, and then calculate the mean value and variance of each $a_i(t)$.
(2) Classify $a_i(t)$ on the basis of whether the results of Z-test deviate from zero significantly. Then, several $a_i(t)$ are divided into high and low frequency components.
(3) Add all the $a_i(t)$ in each component respectively, and get the new time series by composition.

Toward the end, the IMFs and residue are composed into three components: short-term fluctuations caused by normal market activities, the effect of extreme events and a long-term trend.

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Overall process of forecasting methods for BDI

In this section, two forecasting methods for BDI are proposed. The first method is based on EMD and ANN; the second method is based on EEMD (EMD with composition process) and ANN.

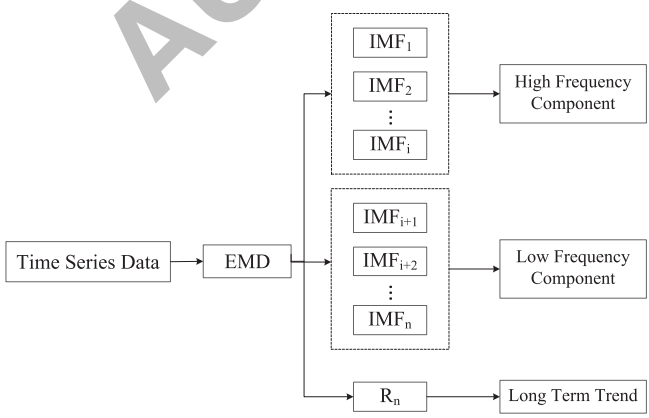


Figure 1: The process of improved EMD method.

Forecasting method based on EMD and ANN

Let $x(t)$, $t = 1, 2, \ldots, N$ denote the BDI time series. The proposed forecasting method can be illustrated as in Figure 2. The method is composed of the following three main steps:

Step 1: Decompose the original time series $x(t)$, $t = 1, 2, \ldots, N$ into n IMFs and one residue R_n by EMD.

Step 2: A three-layer FNN is used as forecasting method to model the IMFs and residue, and makes the corresponding prediction for each component.

Step 3: The prediction results for the IMFs and residue generated by FNN in Step 2 are combined, and the final prediction results are obtained.

Forecasting method based on EEMD and ANN

The EEMD and ANN method includes the step of composition. ANN is used to model each of the CCs and makes the corresponding prediction. Details of the process are given in Figure 3.

Step 1: Decompose the original time series $x(t)$, $t = 1, 2, \ldots, N$ into n IMFs and one residue by EMD.

Step 2: By the composition process, the IMFs and residue are reconstituted to three components, namely, high frequency, low frequency and long-term trend series.

Step 3: FNN is used to model the three components and makes the corresponding prediction for each component.

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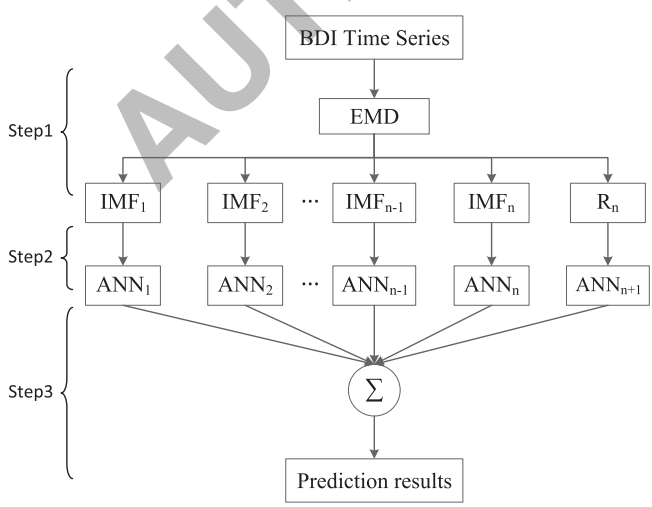


Figure 2: The proposed EMD-ANN forecasting method for BDI time series.

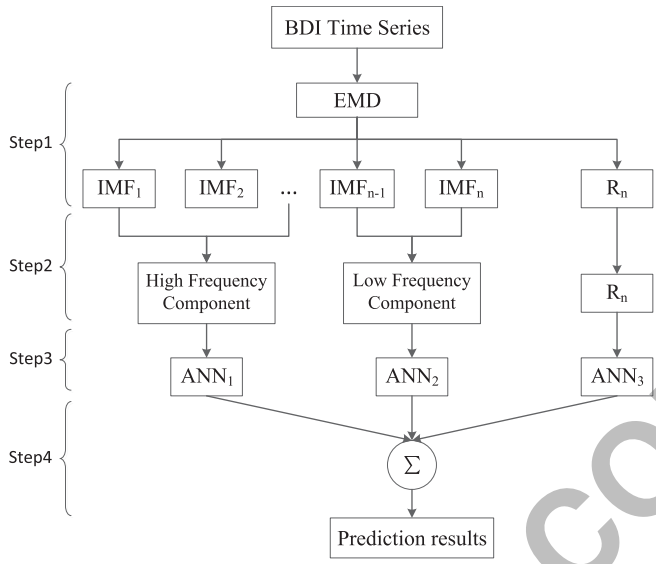


Figure 3: The proposed EEMD-ANN forecasting method for BDI time series.

Step 4: The prediction results for the three components generated by FNN in the previous step are combined, and the final prediction results for BDI are obtained.

The Empirical Experiment

Data sets

To meet the requirements of participants of the dry bulk shipping market, the BFI is published since 1985 by the London Baltic Exchange. Since 1 November 1999, BDI has replaced BFI. Presently, BDI is widely used by industry practitioners and it is regarded as the 'barometer' of the dry bulk shipping market. In this article, the BDI from 1 November 1999 to 29 November 2013 is selected as the experimental data for continuity (Figure 4).

Figure 4 shows that the fluctuation of BDI has the characteristics of large amplitude, high frequency and irregularity. From November 1999 to early 2003, the series fluctuates gently, and market volatility increases from 2003. BDI reaches 11 793 on 20 May 2008. In the following 6 months, it drops sharply to about 700 points.

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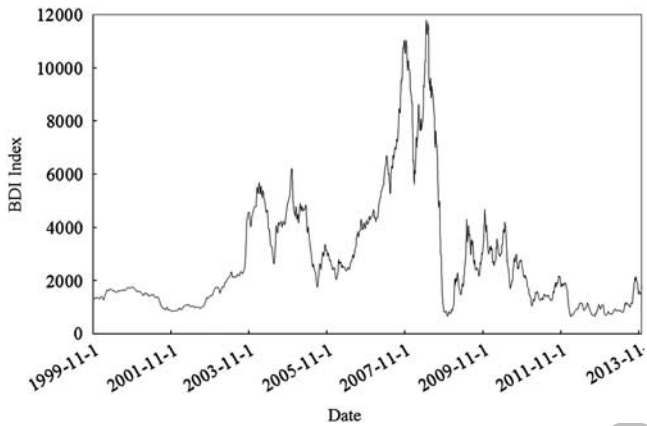


Figure 4: The graph of BDI series.
Source: Clarkson research services limited.

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The non-linear characteristic of BDI can be seen in Figure 4. Furthermore, the unit root test obtained by the Augmented Dickey–Fuller (ADF unit root test) is used to identify the stability of the BDI index. The result is -1.873, larger than -3.432, -2.862 and -2.567, which are the thresholds at 1, 5 and 10 per cent significance level, respectively. This indicates the non-stationarity characteristic of BDI.

Decomposition and composition

First, BDI is decomposed into eight IMFs and one residue by the EMD method. The decomposition results are shown in Figure 5. The IMFs are sorted by fluctuation frequency from high to low. There is almost no fluctuation of residue series, which means that dry bulk freight rates maintain a level of relative stabilization without interference factors.

On the basis of the decomposition results in Figure 5, composition is implemented. The *Z*-test is used to classify the IMFs, and results are shown in Figure 6. The mean value of IMF6 deviates significantly from zero first. Thus, it is regarded as the boundary to classify IMFs, and eight IMFs are composed into two parts. The high frequency component includes the first five IMFs, and the low frequency component includes IMF6 to IMF8.

Therefore, the original BDI time series is composed into three components: high frequency component caused by normal market activities, low frequency component caused by the effects of extreme events and residue representing the

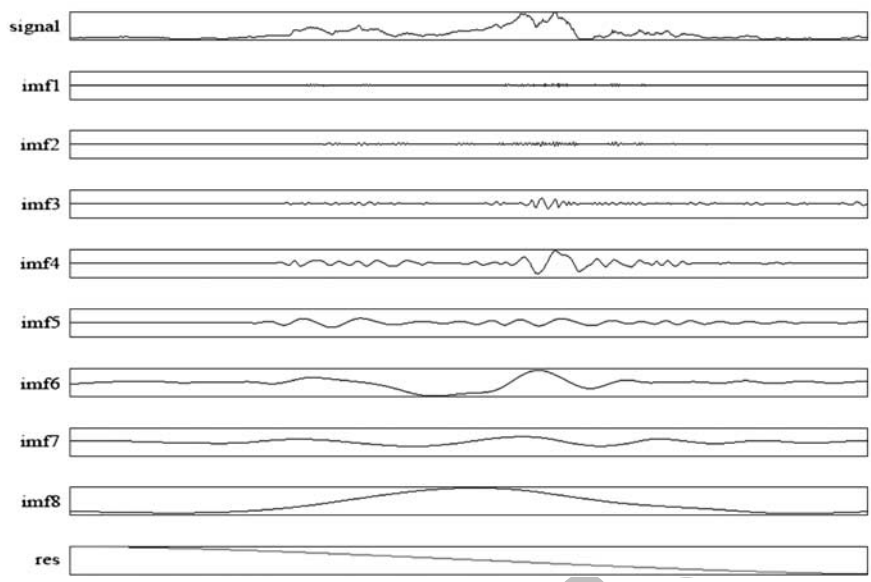


Figure 5: The decomposition of BDI time series (decomposed by authors based on the data of Clarkson Research Services Limited).

long-term trend. Figure 7 shows the graph of the three components and the original BDI time series. The details are illustrated as follows.

Long-term trend component

The long-term trend (residue) of BDI represents the basic dry bulk freight level determined by the long-term supply–demand relationship. Figure 7 demonstrates that the long-term trend stays at around 3800 index points, a level corroborated by industry practitioners. For example, a report published in 2009 by Changjiang Securities showed that if BDI exceeds 3000 points, ship operators are still profitable even after having purchased secondhand ships at prices as high as those of 2008. Meanwhile, information provided by Bloomberg shows that when the BDI exceeds 4000, bulk shipping is in a boom market.

Furthermore, the long-term trend series decreases slightly from 4023 to 3855 during 1999–2013. In spite of bunker price increases (for example, the 380 cst increased from US$152 per ton in 2003 to $700 per ton in July 2008), technical innovations improve the design and efficiency of ships (Chen *et al,* 2012), thus decreasing operational costs. In this way, the long-term equilibrium of the dry bulk market is relatively stable.

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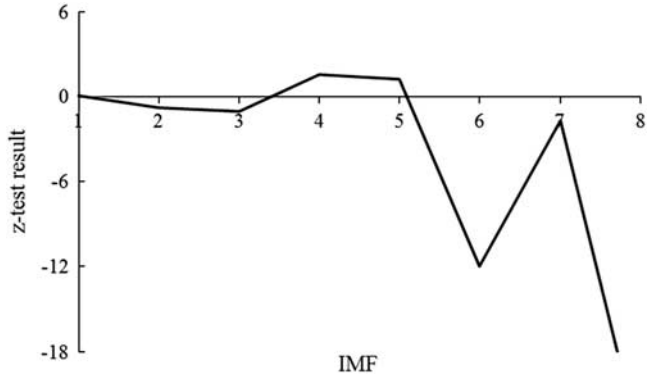


Figure 6: *Z*-test result of IMFs.

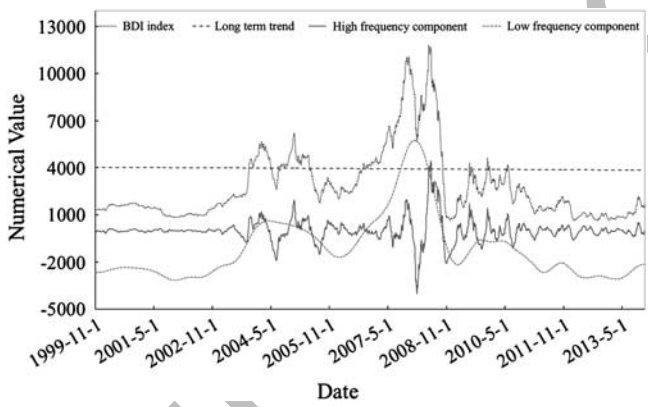


Figure 7: Composition results of IMFs and residue.

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High frequency component

The high frequency component represents normal fluctuations caused by short-term factors, such as variations in international trade, economic or shipping policies, market speculative behavior and expectations of market participants. As shown in Figure 7, the curve of the high frequency component fluctuates around a baseline of market equilibrium.

Since 2002, market participants have shown a strong demand for hedging, which explains the rapid expansion of Forward Freight Agreements (FFA). As can been seen in Figure 7, the line of the high frequency component has significantly irregular short-term fluctuations since 2002. In 2008, countries pursued fiscal policies in succession to stimulate economic recovery. Fluctuations of the high frequency component curve thus increased.

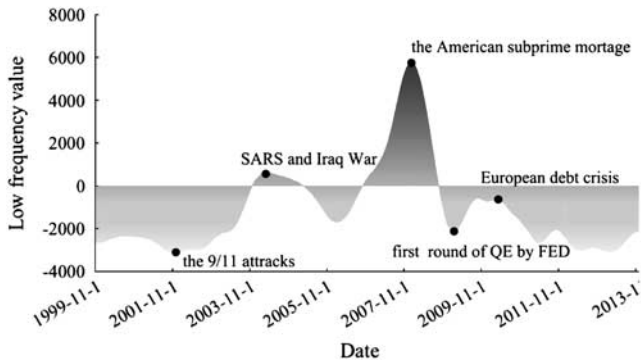


Figure 8: The curve of low frequency component.

Low frequency component

The low frequency component reflects the influence of extreme events, such as epidemic diseases, terrorist attacks, local military conflicts, trade wars, major changes in international relationships and politics, and so on. Detailed effects of extreme events are shown in Figure 8.

As shown in Figure 8, the fluctuation of the low frequency component is slower than that of high frequency one. From 1999 to 2013, the main extreme events influencing dry bulk market included the 9/11 attacks (2001), SARS (2003) and Iraq War (2003), the American subprime mortgage crisis (2007), first round of QE by FED (2008), European debt crisis (2009) and so on. The low frequency series reflects the influence of extreme events efficiently.

Prediction results

In this section, the performances of the four forecasting methods are compared, as follows:

(1) *EMD-ANN*: Decomposing BDI time series into IMFs and a residue; then making predictions for each IMF and the residue by ANN; and obtaining the final prediction results by combining the previous predictions.
(2) *EEMD-ANN*: Decomposing BDI time series into IMFs and a residue, composing them into three components; then making predictions for the CCs by ANN; and obtaining the final prediction results by combining the previous predictions.
(3) *ANN*: Predicting BDI time series directly by ANN model.
(4) *VAR*: Predicting BDI time series by vector auto-regression (VAR) model.

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The first two are the methods proposed in this article, and VAR is seen as a benchmark method. Daily and weekly data of BDI are selected respectively to demonstrate the performance of the four methods.

The correct selection of accuracy metrics is vital to compare performance of various forecasting methods. In this article, four accuracy metrics – MAD (Mean Absolute Deviation), MSE (Mean Square Error), MdRAE (Median Relative Absolute Error) and MAPE (Mean Absolute Percentage Error) – are selected to analyze the forecasting performance of BDI. The selected accuracy metrics and calculations are given in Table 1. MAD and MSE are not scale-free accuracy metrics, thus their scales are determined by the scale of the data. They work well in data sets of the same scale, though they have good description of forecasting accuracy as a traditional accuracy metric. Furthermore, MSE is more sensitive to outliers, which undoubtedly increases the weight of bigger forecast errors. MdRAE is an accuracy metric based on median and relative errors, which achieves scale-free with benchmark method, and could overcome outlier sensitivity. MAPE is a percentage error metric, the greatest advantage of which is scale-free, thus it is appropriate for different scale data sets. However, as a denominator, the actual data has a significant impact on MAPE. In other words, if actual data is close to zero, whatever the forecast error is, MAPE is high.

Daily prediction

The volatility of BDI series has the characteristic of periodicity. To verify the validity and the prediction performance of our four methods in different stages of the shipping cycle, daily data is divided into four stages.

Stage 1: Daily BDI data from 1 November 1999 to 24 December 2002 are selected. Six hundred and fifty-six observations from 1 November 1999 to 14 June 2002 are used for model calibration and 136 observations from 17 June 2002 to 24 December 2002 are used to test prediction performance. Thus, forecasts are

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Table 1: Accuracy metrics and calculations

Accuracy metrics	*Calculation*
MAD	$\frac{1}{n} \times \sum_{i=1}^{n} \lvert y_i - \hat{y}_i \rvert$
MSE	$\frac{1}{n} \times \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$
MdRAE	$median\left(\left\lvert \frac{y_i - \hat{y}_i}{e_i^*} \right\rvert\right)$
MAPE	$\frac{1}{n} \times \sum_{i=1}^{n} \left\lvert \frac{y_i - \hat{y}_i}{y_i} \right\rvert$

Note: n is the number of testing observations, y_i and $\hat{y}_i$ represent actual value and predicted value respectively, e_i^* represent the forecast error of benchmark method.

estimated by the data of a relatively stable period, and tested by the data of a rising period.

Stage 2: Daily BDI data from 5 January 2004 to 22 December 2006 are selected. Six hundred and fourteen observations from 5 January 2004 to 16 June 2006 are used for model calibration and 134 observations from 19 June 2006 to 22 December 2006 are used to test prediction performance. Forecasts are estimated by data of a period of obvious drastic fluctuations, and tested by data of a slightly rising period.

Stage 3: Daily BDI data from 8 January 2007 to 24 December 2009 are selected. Five hundred and ninety-four observations from 8 January 2007 to 22 May 2009 are used for model calibration and 152 observations from 26 May 2009 to 24 December 2009 are used to test prediction performance. Forecasts are estimated by data of a booming period, and tested by the data of a slack period.

Stage 4: Daily BDI data from 4 January 2010 to 21 December 2012 are selected. Five hundred and ninety-nine observations, from 4 January 2010 to 25 May 2012, are used for model calibration and 147 observations, from 28 May 2012 to 21 December 2012, are used to test prediction performance. Forecasts are estimated by data of a period of continuing decline, and tested by the data in a downturn.

Results of the four methods are given in Figure 9 and different indicators of prediction accuracy are given in Table 2. Results in Table 2 demonstrate that EMD-ANN outperforms the other three methods in all four stages. EMD-ANN is based on ANN by decomposing the original time series with EMD before prediction. Compared with ANN, the prediction accuracy of EMD-ANN improves significantly. For example, MAD decreases within a narrow range, the values of MAPE decrease from 0.0021, 0.0040, 0.0135, 0.0098 to 0.0011, 0.0029, 0.0094, 0.0077, respectively, while the value of MSE was reduced nearly 50 per cent in each case. In addition, the indicator of MdRAE, showing the forecast error diminishing sharply without outlier interference, is based on the benchmark VAR model. This indicates that the decomposition of EMD actually helps to improve forecasting accuracy.

Furthermore, EEMD-ANN does not outperform EMD-ANN (as most indicators show), which suggests that the additional composition process of EMD cannot improve prediction performance. This is because each CC is constituted by several IMFs with different volatility frequency. Thus, it is more difficult for ANN to model CCs than IMFs. However, the composed process of EMD helps to retain the original economic nature of the BDI series and understand the factors influencing BDI volatility. This provides a vital technique for dry bulk market analysis.

Weekly prediction

Weekly data ranging from 5 November 1999 to 21 December 2012 are used to illustrate the validity of our four methods in weekly predictions. The total number of observations is 666, in which 535 observations (from 7 January 2000

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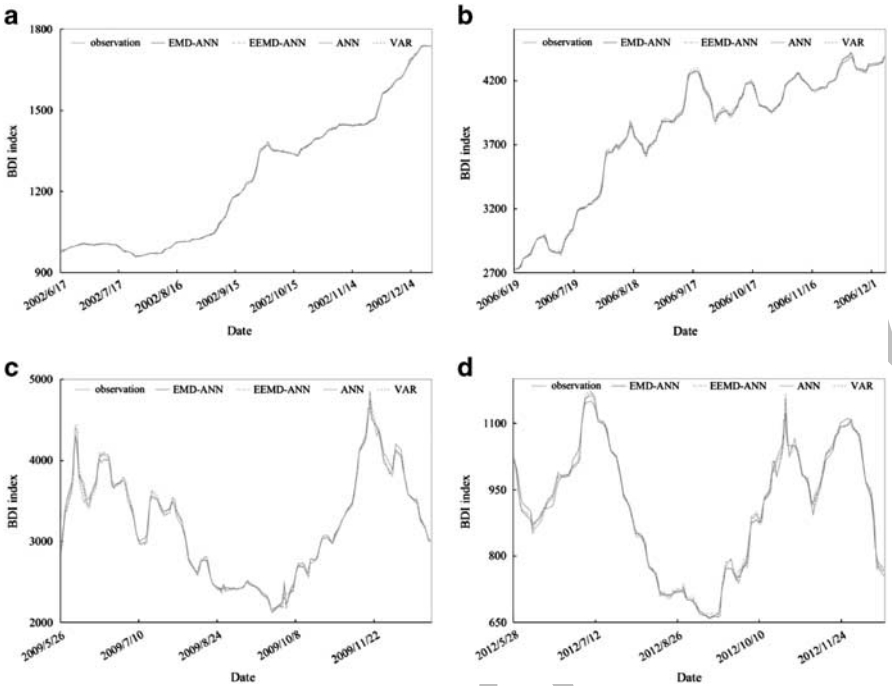


Figure 9: The prediction results. (**a**) Prediction results of stage 1; (**b**) prediction results of Stage 2; (**c**) prediction results of Stage 3; (**d**) prediction results of Stage 4.

to 7 May 2010) are used for model calibration and 131 observations (from 14 May 2010 to 23 November 2012) are used to test prediction performance. Results of the four methods are given in Figure 10 and Table 3.

Similar to the results of daily predictions, EMD-ANN is the best performing method in terms of weekly prediction. Results of Table 3 demonstrate that the indicators of EMD-ANN are less than those of the other three methods. Comparing EMD-ANN, EEMD-ANN and ANN, all of the indicators of EEMD-ANN are undesirable, suggesting that the composition process does not help to improve the weekly prediction of BDI. The composition of IMFs, obtained by decomposition with EMD, would contribute in retaining economic characteristics of original time series, but it has a negative effect on prediction accuracy and predictive effect.

To sum up, EMD-ANN outperforms all other methods in daily and weekly predictions. The decomposition of EMD helps to improve prediction accuracy, but this is not true for the composition process. In addition, the accuracy of daily prediction is higher than that of weekly prediction.

Table 2: Prediction accuracy indicators of daily data in four stages

Stage	*Method*	*Indicators*							
		MAD		*MAPE*		*MSE*		*MdRAE*	
		Value	*Rank*	*Value*	*Rank*	*Value*	*Rank*	*Value*	*Rank*
1	EMD-ANN	1.297	1	0.0011	1	3.909	1	0.406	1
	EEMD-ANN	2.544	2	0.0020	2	13.659	3	0.999	3
	ANN	2.622	3	0.0021	3	13.310	2	0.980	2
	VAR	2.834	4	0.0022	4	16.574	4	1	4
2	EMD-ANN	11.199	1	0.0029	1	210.068	1	0.610	1
	EEMD-ANN	13.360	2	0.0036	2	312.299	2	0.736	2
	ANN	14.805	3	0.0040	3	362.398	3	0.856	3
	VAR	16.282	4	0.0043	4	444.100	4	1	4
3	EMD-ANN	30.330	1	0.0094	1	2066.148	1	0.556	2
	EEMD-ANN	30.403	2	0.0095	2	2190.044	2	0.520	1
	ANN	43.767	3	0.0135	3	4278.919	3	0.919	3
	VAR	47.732	4	0.0146	4	4916.221	4	1	4
4	EMD-ANN	7.177	1	0.0077	1	101.750	1	0.691	1
	EEMD-ANN	7.198	2	0.0077	1	102.365	2	0.811	2
	ANN	8.977	3	0.0098	3	192.251	3	0.913	3
	VAR	9.191	4	0.0100	4	204.070	4	1	4

Notes: MAD, MAPE, MSE and MdRAE are Mean Absolute Deviation, Mean Absolute Percentage Error, Mean Square Error and Median Relative Absolute Error, respectively.

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Prediction verification of various routes

To further verify the validity of the proposed method, the freight rate series of four routes are selected. Considering that the freight index of various routes is reported by Clarkson from 1 November 2002, daily data of the four routes, from 1 November 2002 to 21 December 2012, are selected. A total of 1998 observations (from 1 November 2002 to 29 October 2010) are used for model calibration and 536 observations (from 1 November 2010 to 21 December 2012) are used to test prediction performance. Results are shown in Table 4.

Table 4 compares the prediction accuracy indicators of EMD-ANN, EEMD-ANN, single ANN and VAR. Results demonstrate that the EMD-ANN method performs better than VAR and ANN, confirming the effectiveness of EMD in improving prediction precision of freight index. Furthermore, ANN and VAR models have almost similar prediction accuracy.

In this section, the results of VAR are used as benchmark to calculate the indicator of MdRAE. The MdRAE of ANN model for all routes are less than 1 with little difference, indicating that ANN does not improve prediction accuracy obviously compared to VAR. The MdRAE values of EMD-ANN method are far below 1, suggesting that EMD-ANN improves prediction accuracy obviously. However, not all the MdRAE values of EEMD-ANN are lower than 1, which

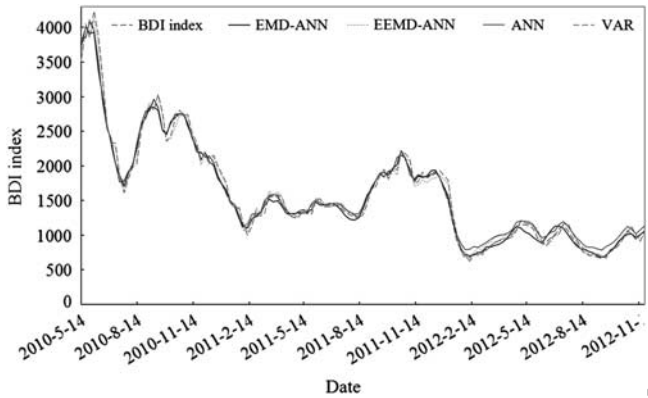


Figure 10: Prediction results of weekly data.

Table 3: Prediction accuracy indicators of weekly data

Model	*Indicators*							
	MAD		*MAPE*		*MSE*		*MdRAE*	
	Value	*Rank*	*Value*	*Rank*	*Value*	*Rank*	*Value*	*Rank*
EMD-ANN	41.986	1	0.029	1	3045.02	1	0.546	1
EEMD-ANN	76.020	3	0.048	3	10 359.3	3	0.927	3
ANN	42.690	2	0.044	2	3049.05	2	0.656	2
VAR	80.778	4	0.049	4	12 755.3	4	1	4

Notes: MAD, MAPE, MSE and MdRAE are Mean Absolute Deviation, Mean Absolute Percentage Error, Mean Square Error and Median Relative Absolute Error, respectively.

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indicates that the composed process of decomposition would affect the validity of EMD in improving prediction accuracy. However, the composed process retains the economic nature of the original BDI series, which is lost during the process of decomposition.

Conclusions

Dry bulk shipping has the characteristics of high volatility and cyclicality. In this article, a forecasting method for BDI, based on EMD and ANN, was proposed. Two combined forecasting frameworks were developed, namely, 'decomposition-prediction' (EMD-ANN) and 'decomposition-composition-prediction' (EEMD-ANN). Empirical experiments were provided to illustrate the

Table 4: Prediction accuracy indicators of various routes

Route	*Method*	*Indicators*							
		MAD		*MAPE*		*MSE*		*MdRAE*	
		Value	*Rank*	*Value*	*Rank*	*Value*	*Rank*	*Value*	*Rank*
BPI P1A_03	EMD-ANN	91	1	0.0099	1	19 007.0	1	0.712	1
	EEMD-ANN	128	3	0.0135	3	43 456.1	4	0.943	2
	ANN	127	2	0.0132	2	40 402.2	2	0.952	3
	VAR	132	4	0.0140	4	43 405.7	3	1	4
BPI P2A_03	EMD-ANN	83	1	0.00425	1	16 395.8	1	0.577	1
	EEMD-ANN	120	3	0.00593	3	29 872.8	3	1.001	4
	ANN	115	2	0.00569	2	28 839.5	2	0.948	2
	VAR	122	4	0.00604	4	31 881.0	4	1	3
BCI C8_03	EMD-ANN	502	1	0.0223	1	635 568	1	0.612	1
	EEMD-ANN	823	3	0.0359	4	1 703 527	2	0.987	2
	ANN	819	2	0.0353	2	1 738 187	3	0.998	3
	VAR	827	4	0.0359	3	1 862 926	4	1	4
BCI C9_03	EMD-ANN	264	1	0.0086	1	163 491	1	0.510	1
	EEMD-ANN	471	3	0.0158	4	551 981	2	1.020	4
	ANN	466	2	0.0154	2	603 185	3	0.979	2
	VAR	474	4	0.0157	3	606 904	4	1	3

Notes: MAD, MAPE, MSE and MdRAE are Mean Absolute Deviation, Mean Absolute Percentage Error, Mean Square Error and Median Relative Absolute Error, respectively.

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validity of the proposed method. Our results indicate that EMD-ANN outperforms EEMD-ANN and other two methods (ANN and VAR).

The main contribution of this article is its novel method, as well as an effective approach for analysis and prediction of dry bulk freight rates. The proposed method can efficiently tackle the non-linear and non-stationary characteristics of the BDI series, and improve prediction accuracy. It provides an effective approach for dry bulk market analysis and is meaningful for market participants to grasp market dynamics. Furthermore, this article contributes to *forecasting research*. The proposed method not only integrates EMD with ANN, but incorporates a composition process. The interesting finding is that the decomposition process helps to improve prediction performance while this not true for the composition process. However, the latter process contributes in retaining the economic characteristics of the original time series, and it provides an important approach for dry bulk market analysis. The proposed method and our findings help to extend forecasting research and potentially provide research opportunities in related fields.

In this article, the final BDI prediction results were obtained by accumulation of all prediction results of IMFs and residue, while the inherent relationships among IMFs were not considered. Thus, this article serves as an ideal platform for further research to analyze the relationship of IMFs and improve prediction accuracy. Furthermore, BDI includes several subindexes of different ship types, such as BCI, BPI and BHI. Therefore, more empirical studies for freight rate series of different ship types are possible so as to verify the reliability and applicability of the proposed method.

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